

Student Performance Classification Using K-Nearest Neighbours

Team members

Zayed khan (23-51421-1)

Ashraful Hossain (23-50722-1)

Nanda Gopal Das (23-50703-1)

1. Problem Statement

Student academic performance is influenced by several factors, including study time, previous failures, absences, family background, school-related factors, and other personal and social characteristics. Identifying students who may be at risk of failing can help educational institutions provide support before the final examination. The objective of this project is to develop a binary classification system that predicts whether a student is likely to Pass or Fail based on available student information. The project uses the UCI Student Performance dataset, specifically the Mathematics dataset. The final grade (G3) is used to create the target variable:

- Pass (1): $G3 \geq 10$
- Fail (0): $G3 < 10$

The previous-period grades G1 and G2, as well as the final grade G3, are excluded from the input features. This is done to avoid making the prediction artificially easy and to investigate whether student characteristics available before the final grade can be useful for identifying academic risk.

2. Stakeholders

The main stakeholders of this project are:

Students

Students can benefit from early identification of academic difficulties. A prediction system could help them understand when additional academic support may be necessary.

Teachers

Teachers can use the prediction results as an additional indication of which students may require more attention, guidance, or academic support.

Educational Institutions

Schools and educational administrators can use such a system to understand student performance patterns and potentially develop early-intervention programs.

Academic Researchers

Researchers can use the project as an example of applying machine-learning classification techniques to educational data.

3. Objective

The main objective of this project is to develop and evaluate a machine-learning classification system for predicting student academic performance.

The specific objectives are:

1. To analyze the UCI Student Performance dataset.
2. To understand the distribution and relationships among student-related variables.
3. To create a binary Pass/Fail target variable using the final grade.
4. To identify useful patterns and relationships through exploratory data analysis.
5. To divide the dataset into training and testing sets while maintaining the class distribution.
6. To apply appropriate preprocessing techniques to numerical and categorical variables.
7. To establish a simple baseline model for comparison.
8. To develop a K-Nearest Neighbours (KNN) classification model.
9. To compare KNN with other candidate classification models.
10. To evaluate the models using appropriate classification metrics.
11. To analyze model errors and discuss the limitations and responsible use of the prediction system.

4. Success Criteria

The project will be considered successful if it satisfies the following criteria:

- The complete machine-learning workflow can be executed successfully.
- The dataset is properly inspected and documented.
- Missing values, duplicate records, data types, and possible data-quality issues are investigated.
- The train-test split is performed correctly without data leakage.
- Preprocessing is fitted only on the training data.
- A simple baseline model is established.
- At least two candidate machine-learning models are developed and compared.

- KNN is properly scaled because it is a distance-based algorithm.
- Model performance is evaluated using metrics such as:
 - Accuracy
 - Precision
 - Recall
 - F1-score
 - ROC-AUC where appropriate
- Confusion matrices are used to understand classification errors.
- The final test set remains untouched during model selection and tuning.
- Results are reproducible using a fixed random state.
- The project provides practical interpretation rather than relying only on a high accuracy score.

5. Dataset Source

The dataset used in this project is the Student Performance dataset from the UCI Machine Learning Repository.

Dataset: Student Performance

Repository: UCI Machine Learning Repository

Number of records: 395 students

Number of attributes: 33

Authors: Paulo Cortez and related contributors

The dataset contains information about students from Portuguese secondary schools. It includes academic, demographic, social, family, and school-related attributes.

The dataset contains variables such as:

The final grade G3 is used to create the classification target. However, G1 and G2 are excluded from the model features because they are strongly related to G3 and could make the prediction unrealistically easy.

6. Distribution and Relationship Analysis

Exploratory Data Analysis (EDA) was performed to understand the distribution of the target variable and the relationships between important student characteristics and academic performance.

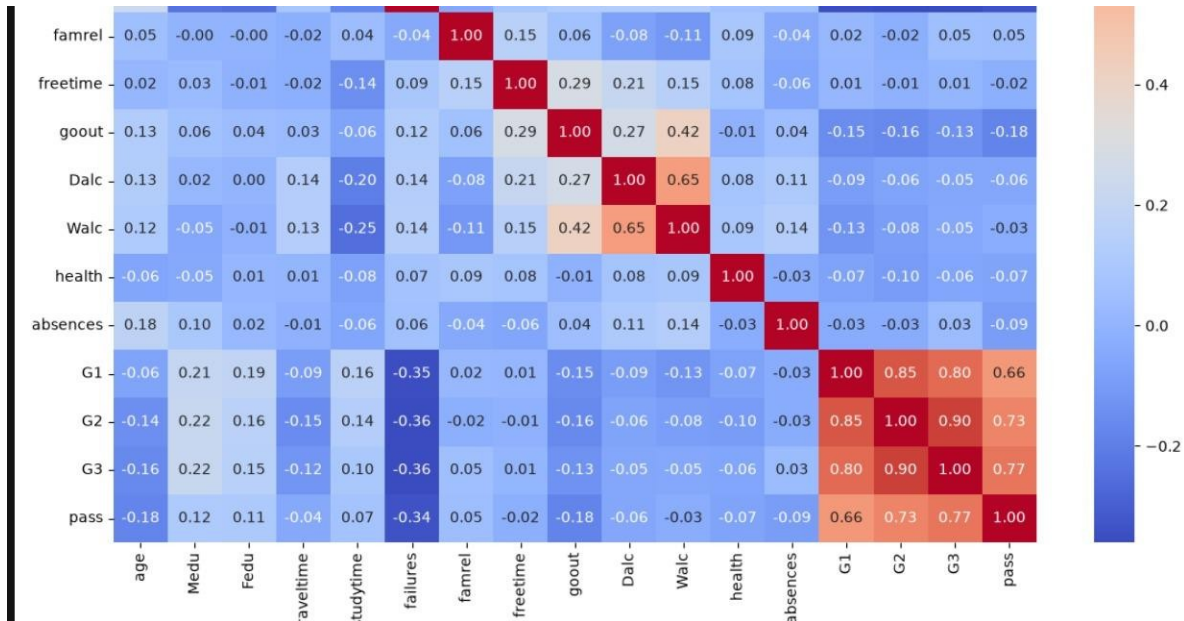
The graphs will be added to the report later.

6.1 Pass/Fail Distribution

The target variable was created using:

Students with a final grade of 10 or above are classified as Pass, while students with a final grade below 10 are classified as Fail.

Figure 1: Correlation graph



The heatmap provides an overview of relationships between numerical variables. It can help identify variables that have stronger or weaker linear associations with the final grade. Correlation is used as an exploratory tool and does not establish causation.

6.3 Study Time and Performance

Study time was examined in relation to the Pass/Fail target.

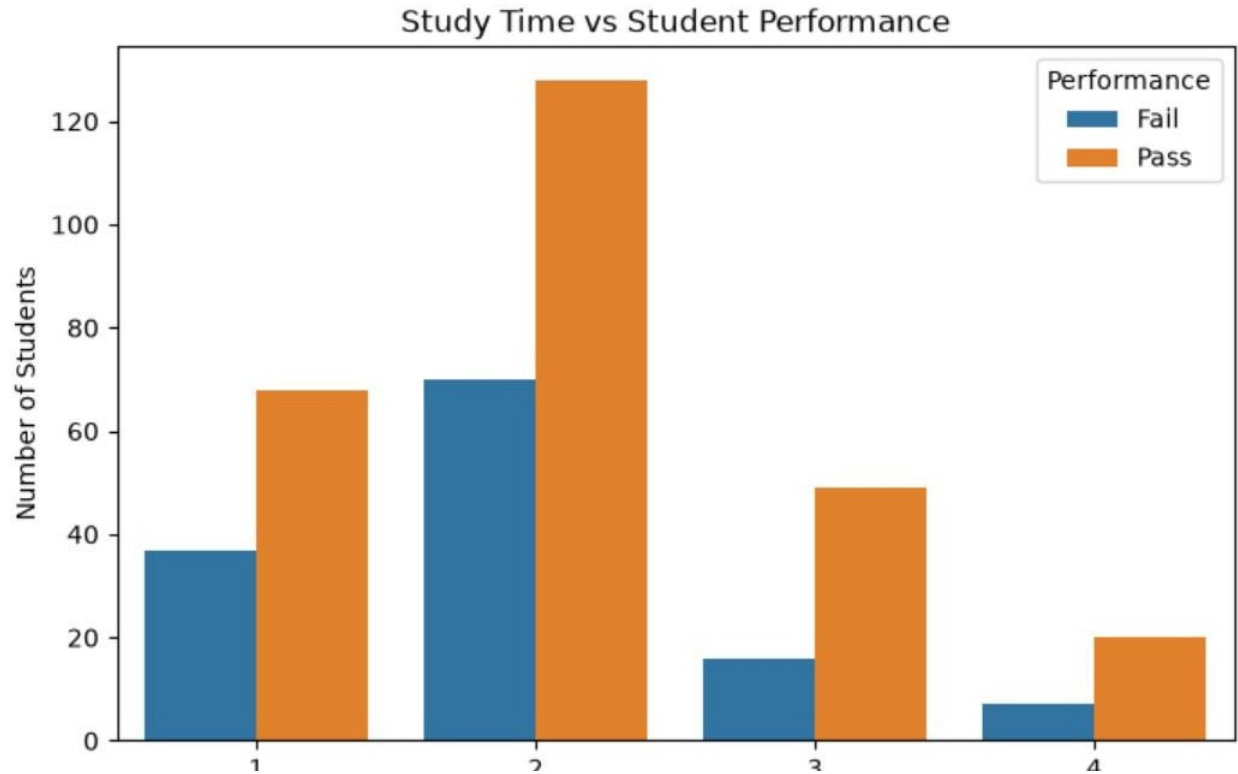


Figure 3: Study Time vs Pass/Fail

This analysis helps investigate whether students with different levels of weekly study time show different patterns of academic performance.

6.4 Absences and Final Grade

The relationship between the number of absences and final grade was examined using a scatter plot.

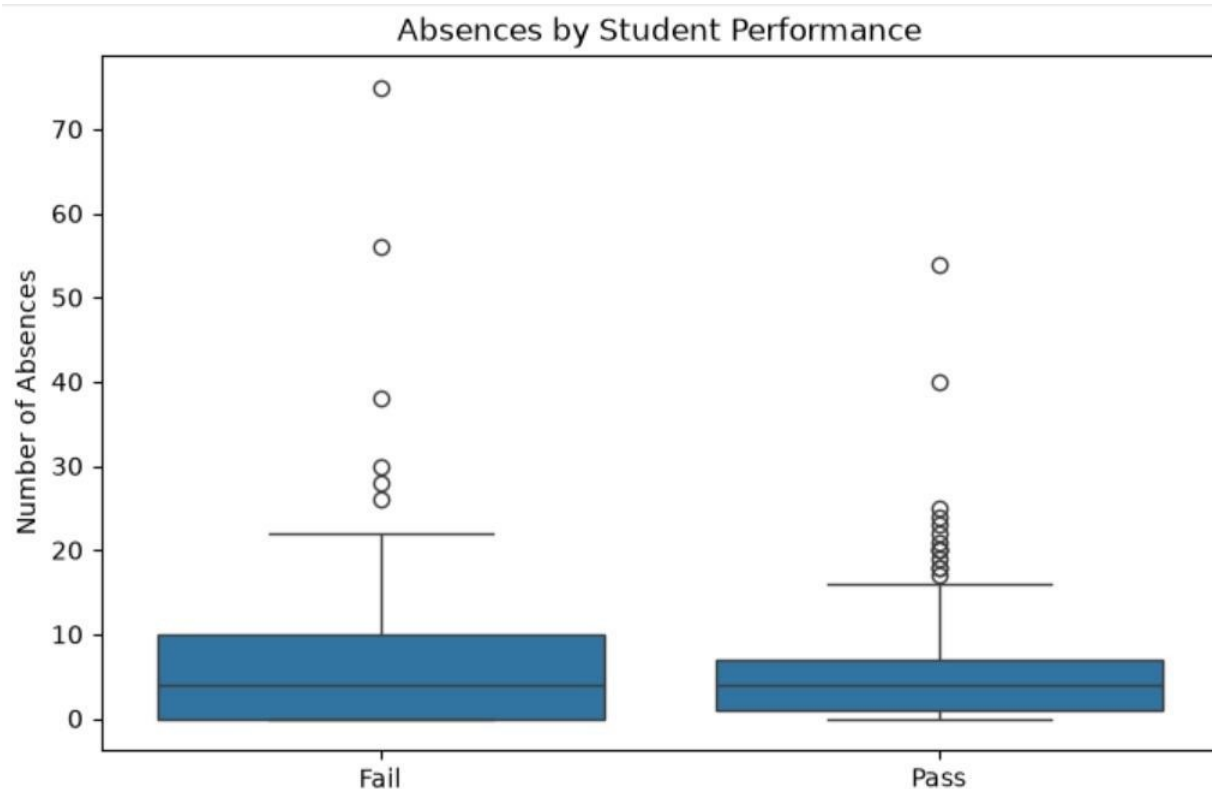


Figure 4: Absences vs Final Grade

7. Train-Test Split

The dataset was divided into two parts:

- 80% training data
- 20% testing data

A stratified split was used to maintain a similar Pass/Fail class distribution in both datasets. The random state was fixed at **42** to make the experiment reproducible. The test dataset was kept separate from the model-development process. It was not used for hyperparameter tuning or model selection. This design provides a more reliable estimate of how the final model performs on previously unseen student records.

8. Pipeline Processing

The dataset contains both numerical and categorical variables. Therefore, different preprocessing techniques are required.

Numerical Features

Numerical variables are processed using:

1. Median imputation for missing values.
2. StandardScaler for feature scaling.

Scaling is particularly important for KNN because KNN calculates distances between observations. Without scaling, variables with larger numerical ranges could have a greater influence on the distance calculation.

Categorical Features

Categorical variables are processed using:

1. Most-frequent-value imputation.
2. One-hot encoding.

`handle_unknown="ignore"` is used in the encoder so that unseen categories in the test data do not cause errors.

A ColumnTransformer is used to apply the correct preprocessing to each feature type.

The preprocessing and model are combined using a Pipeline.

Conceptually, the pipeline follows:

Raw Data



Train/Test Split



Preprocessing

├── Numerical → Imputation → Scaling

└── Categorical → Imputation → One-Hot Encoding



Machine Learning Model



Prediction



Evaluation

The pipeline also helps prevent data leakage because preprocessing parameters are learned from the training data rather than from the complete dataset.

9. Baseline Definition and Result

A baseline model provides a simple reference point against which more advanced models can be compared. For this project, a DummyClassifier using the majority-class strategy is used as the baseline. The DummyClassifier does not learn meaningful relationships between student characteristics and academic performance. Instead, it predicts the most common class.

Baseline Result

The baseline model's actual results should be reported using the same evaluation metrics used for the other models.

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
DummyClassifier	[0.679]	[0.679]	[1.00]	[0.8020]	[0.500]

Interpretation:

The baseline provides a reference for determining whether the machine-learning models are learning useful patterns. A candidate model should demonstrate meaningful improvement over this baseline rather than simply achieving a high accuracy score.

10. Models

Several classification models are considered in this project.

10.1 DummyClassifier

The DummyClassifier is used as the baseline. It provides a simple reference performance and does not attempt to learn relationships between the features and target.

10.2 Logistic Regression

Logistic Regression is used as a simple supervised classification model. It is useful as a benchmark because it provides a relatively simple and interpretable classification approach. The model can help determine whether a straightforward linear decision boundary can perform reasonably well on this dataset.

10.3 K-Nearest Neighbours

K-Nearest Neighbours is the main model of this project. KNN classifies a new student by examining nearby training observations and assigning a class based on the classes of its nearest

neighbours.

Important KNN parameters include:

- `n_neighbors`
- `weights`
- `metric`

Different values of `k` can be evaluated to determine which setting provides the best validation performance. Because KNN is distance-based, feature scaling is an important part of the preprocessing pipeline.

10.4 Random Forest

Random Forest is included as a nonlinear comparison model. It combines multiple decision trees and can capture more complex relationships between variables. Comparing Random Forest with KNN and Logistic Regression allows the project to evaluate whether a distance-based model performs competitively against both a simpler linear model and a nonlinear tree-based model.

11. Model Development and Validation

The models are developed using only the training dataset.

For KNN, different hyperparameter combinations can be tested using cross-validation.

A 5-fold stratified cross-validation strategy is used for model selection.

The final test set is not used during this tuning process.

The main optimization metric is the F1-score, because it considers both precision and recall.

12. Error Analysis

After selecting the final model, its predictions on the untouched test set are analyzed.

Figure 8: Confusion Matrix

False negatives are particularly important in this project. A false negative occurs when a student who actually fails is predicted as passing. If the system were used for early academic intervention, such a prediction could cause a student who needs additional support to be overlooked. False positives also matter because a passing student may be incorrectly identified as at risk. However, this may be less harmful if the prediction is used only as a screening tool and not as a final decision. The model should therefore not be treated as a replacement for teachers or academic advisors.

13. Conclusion

This project developed a machine-learning classification workflow for predicting whether a student is likely to pass or fail based on student-related information.

The UCI Student Performance dataset was analyzed through data auditing and exploratory data analysis. Several distributions and relationships were investigated, including study time, absences, previous failures, age, and other student characteristics.

A proper train-test split was used to separate model development from final evaluation. Numerical and categorical variables were processed using an appropriate preprocessing pipeline, including imputation, scaling, and one-hot encoding. This approach also helped reduce the risk of data leakage.

A DummyClassifier was established as a baseline, while Logistic Regression, KNN, and Random Forest were considered as candidate models. KNN was the main focus because the project specifically investigates a distance-based classification approach and demonstrates the importance of feature scaling. The final model should be selected based on its performance on the untouched test set, with particular attention to F1-score, precision, recall, and the confusion matrix. The model's errors should also be considered because incorrect predictions may affect how students are identified for academic support. Overall, this project demonstrates an end-to-end machine-learning workflow, from problem formulation and data analysis to preprocessing, model development, evaluation, and interpretation. The resulting system should be viewed as a supportive analytical tool rather than an automated decision-making system for students.